

Elias Krook

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PROFILE

MSc student in Engineering Mechanics, Solid Mechanics track, at KTH Royal Institute of Technology. Background in product development, workshop/prototyping engineering, and AV/sound engineering, with hands-on experience in CAD, FEM simulation, and prototype manufacturing. Combines analytical work with practical problem-solving in a workshop environment.

EDUCATION

MSc, Engineering Mechanics – Solid Mechanics 2025 – ongoing
KTH Royal Institute of Technology

BSc, Industrial Design Engineering (Design and Product Development) 2023 – 2025
KTH Royal Institute of Technology

EXPERIENCE

Prototype Center Assistant May 2023 – ongoing
KTH ITM School

- Maintenance and repair of workshop machinery
- Production of orders via waterjet cutting
- Conducted safety briefings for workshop visitors
- Laser cutting, 3D printing, and milling

Sketch Assistant May 2023 – ongoing
KTH ITM School

- Supported modules within product development courses, focused on pen-and-paper sketching techniques for fast, clear concept visualization
- Presented exercise assignments to students and graded/corrected submissions

Site Manager Aug 2025 – Jun 2026
Älvsjö AIK Simförening

- Held staff, safety, and sales responsibility for the facility
- Acted as point of contact for members, staff, and partners
- Oversaw daily operations of the swimming facility

Sound Engineer (B-Sound) Aug 2022 – Apr 2023
Kulturhuset Stadsteatern

- Worked in the B-sound (assistant sound engineer) role for theatre performances and productions

Sound / AV Engineer 2017 – 2020
Svensk Kulturfilm Konferensteknik AB

- Conference and event technician focused on live-mixing, lighting, and video routing
- Worked across both private and public sector clients

RELEVANT PROJECTS

Styrka i Stormen — Bachelor Thesis (MF132X) 2025

Developed an outdoor, weather-resistant strength-training rig in collaboration with Recoil/Gymkompaniet. The project included a market and needs study and concept development of three candidate concepts, from which the final concept, E-ATX — a modular half-rack solution with internally routed cables and two integrated weight stacks — was selected based on user needs, technical feasibility, and brand strategy.

Diving Board Dimensioning — FEM Modelling Final Project (SE2860) 2026

Analytical and numerical FEM analysis (COMSOL) of a diving board's structural strength under multiple load cases, including material optimization and comparison between aluminum, plywood, and composite.

FEM Simulation Coursework (SE2860) 2025 – 2026

Course projects in COMSOL: crash-zone analysis, a bicycle damper, an electromagnetic solenoid plunger, and thermo-mechanical analysis of a bimetallic strip/thermostat, each compared against analytical solutions.

Room Acoustics of a Residential Apartment (SD2175) 2026

COMSOL-based simulation and analysis of acoustic conditions in a residential apartment, focused on sound propagation and room modes.

SKILLS

Technical tools: Matlab · COMSOL · Ansys · Solid Edge · AutoCAD · LaTeX

Manufacturing: Additive manufacturing (LPBF & related) · Laser cutting · Waterjet cutting · CNC milling

Other: Research methodology · Technical writing · PowerPoint · Photoshop/Illustrator · Driving licence B (manual)

LANGUAGES

Swedish · English

LINKS

Portfolio: <https://eliaskrook.github.io/Portfolio/> · LinkedIn: [linkedin.com/in/elias-krook-969191127](https://www.linkedin.com/in/elias-krook-969191127)